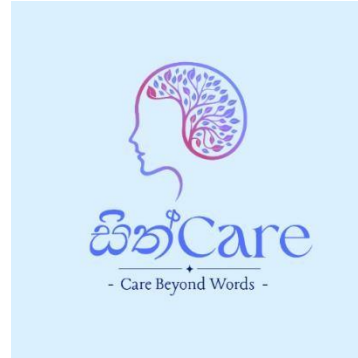




SENG 31242 – System Design Project

SithCare – Mental Health Channelling App

Use Case Lists, Descriptions and Diagrams

Group 10

SE/2022/046 - Lochithya Hettiarachchi

SE/2022/005 - Anushka Dilinuwan

SE/2022/010 - Adhithya Jayawardhana

SE/2022/025 - Vindya Wickramathilaka

SE/2022/041 - Ramesh Krishan

Year 3 Semester 1

Bachelor of Science (Hons.) in Software Engineering

Faculty of Science – Software Engineering Teaching Unit

University of Kelaniya, Sri Lanka

Supervisor – Eng. Dr. Tiroshan Madushanka

24/05/2026



Table of Contents

1. Use Case List	1
2. Use Case Descriptions and Diagrams	2
2.1. UC-01 – Register and Login	2
2.2. UC-02 – Manage User Profile and Consent	4
2.3. UC-03 – Search Mental Health Specialist.....	6
2.4. UC-04 – Book Appointment	8
2.5. UC-05 – Manage Doctor Availability.....	10
2.6. UC-06 – Approve / Reschedule Appointment.....	12
2.7. UC-07 – Join Telemedicine Consultation	14
2.8. UC-08 – Process Payment and Generate Receipt.....	15
2.9. UC-09 – Track Live Patient Queue.....	17
2.10. UC-10 – Request Emergency Support.....	18
2.11. UC-11 – Use Anonymous Community Consultation	20
2.12. UC-12 – Complete AI Triage Questionnaire (SithBot)	22

List of Figures

Figure 1 - UCD 01 (Register and Log in)	3
Figure 2 - UCD 02 (Manage User Profile and Content)	5
Figure 3 - UCD 03 (Search Mental Health Specialist).....	7
Figure 4 - UCD 04 (Book Appointment)	9
Figure 5 - UCD 05 (Manage Doctor Availability)	11
Figure 6 - UCD 06 (Approve/Reschedule Appointment).....	13
Figure 7 - UCD 07 (Join Telemedicine Consultation).....	15
Figure 8 - UCD 08 (Process Payment & Generate Receipt)	16
Figure 9 - UCD 09 (Track Live Patient Queue).....	18
Figure 10 - UCD 10 (Request Emergency Support)	20
Figure 11 - UCD 11 (Use Anonymous Community Consultation)	21
Figure 12 - UCD 12 (Complete AI Triage Questionnaire).....	23



1. Use Case List

ID	Use Case Name	Primary Actor(s)	Description
UC-01	Register and Login	Patient / Doctor / Admin / Volunteer	User creates account with OTP verification, logs in securely, and is routed to their role-based dashboard.
UC-02	Manage User Profile and Consent	Patient	Patient updates personal information, sets language preferences, manages privacy consent, and adds emergency contacts.
UC-03	Search Mental Health Specialist	Patient	Patient searches and filters mental health professionals by category, language, consultation mode, location, and availability.
UC-04	Book Appointment	Patient / Clinic Admin / Doctor	Patient selects a slot and books an appointment; system confirms the booking and notifies all relevant parties.
UC-05	Manage Doctor Availability	Doctor / Clinic Admin	Doctor or clinic admin defines, updates, or blocks available time slots across one or more clinic locations.
UC-06	Approve / Reschedule Appointment	Clinic Admin / Doctor	Clinic staff reviews pending appointment requests and approves, rejects, or reschedules as required.
UC-07	Join Telemedicine Consultation	Patient / Doctor	Patient joins a secure video consultation via a doctor-generated room link; fallback phone call provided if video fails.
UC-08	Process Payment and Generate Receipt	Patient / Payment Gateway / Clinic Admin	Patient pays online (PayHere) or via cash; system updates payment status and generates a PDF receipt.
UC-09	Track Live Patient Queue	Patient / Clinic Admin	System displays the current token being served and the patient's position; clinic admin advances the queue.
UC-10	Request Emergency Support	Patient / Doctor / Emergency Helpline	Patient activates emergency button; system displays crisis options, initiates contact, and shares location if consented.



UC-11	Use Anonymous Community Consultation	Patient / Community Moderator	Patient posts general mental health concerns anonymously; community responds and moderator oversees safety.
UC-12	Complete AI Triage Questionnaire	Patient / AI Triage Module (SithBot)	Patient answers guided questions; SithBot recommends a specialist category with a mandatory non-diagnosis disclaimer.

2. Use Case Descriptions and Diagrams

2.1. UC-01 – Register and Login

Use Case ID	UC-01
Use Case Name	Register and Login
Actor(s)	Patient, Doctor, Clinic Admin, Volunteer, Platform Admin
Preconditions	User is not logged in. User has a valid email address or phone number.
Postconditions – Success	Authenticated session created; user routed to correct role-based dashboard.
Postconditions – Failure	Account not created or login rejected; appropriate error message displayed.
Main Flow	1. User opens SithCare and navigates to /register or /login.
	2. For registration: user selects role, enters full name, email, phone, date of birth, preferred language, and password. 3. System validates input (email format, password strength, phone format). 4. System checks for duplicate email/phone. 5. System sends OTP to registered email/phone. 6. User enters OTP, System verifies OTP, activates account, and redirects to profile completion wizard. 7. For login: user enters credentials, system verifies role and account status, creates a secure JWT session, and redirects to dashboard.
Alternative Flow 1	At step 2 (Registration): If password is weak, system highlights requirements (min 8 chars, 1 uppercase, 1 digit, 1 special character). User corrects and resubmits.



Alternative Flow 2	At step 5: If email is already registered, system displays a generic message to prevent user enumeration. OTP resend is limited to 3 attempts per hour.
Alternative Flow 3	At step 8 (Login): Invalid credentials or suspended account – system displays error. After 5 failed attempts, account is temporarily locked for 15 minutes.
Business Rules	- BR-01: OTPs expire after 10 minutes. - BR-02: Max 3 OTP resend attempts per hour. - BR-03: Min password: 8 chars, 1 uppercase, 1 digit, 1 special character. - BR-04: JWT access tokens expire after 15 minutes; refresh tokens expire after 7 days.

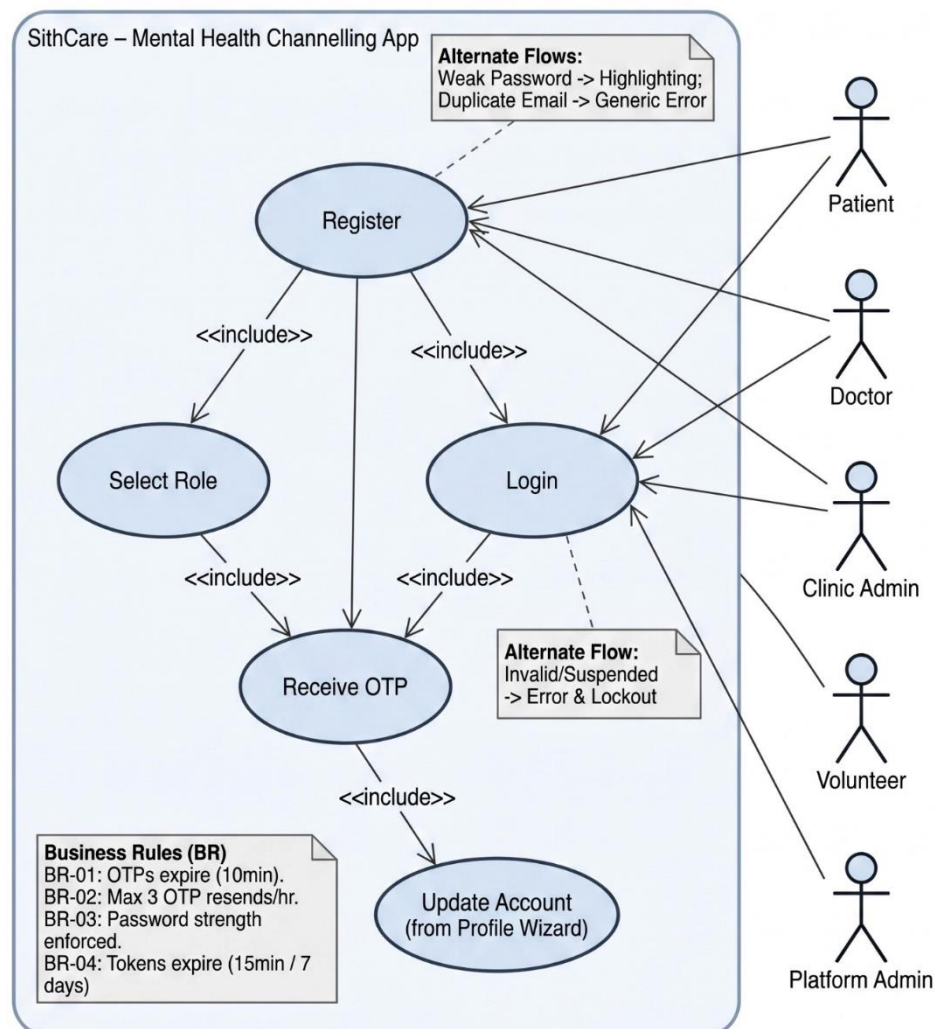


Figure 1 - UCD 01 (Register and Log in)



2.2. UC-02 – Manage User Profile and Consent

Use Case ID	UC-02
Use Case Name	Manage User Profile and Consent
Actor(s)	Patient
Preconditions	Patient is registered and logged in to their dashboard.
Postconditions – Success	Patient profile is updated with the latest personal information, consent preferences, and emergency contact details.
Postconditions – Failure	Profile changes not saved; validation error message displayed; existing data remains unchanged.
Main Flow	1. Patient navigates to 'My Profile' from the dashboard. 2. Patient views current profile information: name, contact details, date of birth, preferred language, and profile photo. 3. Patient edits desired fields and clicks 'Save Changes'. 4. System validates inputs (e.g., phone number format, valid date of birth). 5. System saves updated profile and displays a success confirmation. 6. Patient navigates to 'Privacy & Consent' settings.
	7. System displays privacy policy and current consent status. 8. Patient reviews and updates consent preferences (e.g., data sharing for anonymous analytics). 9. Patient adds or updates emergency contact name and phone number. 10. System saves preferences and logs the consent update with a timestamp.
Alternative Flow 1	At step 4: If validation fails (e.g., invalid phone format), system highlights the specific field error and prevents saving until corrected.
Alternative Flow 2	At step 8: If patient withdraws previously given consent, system updates settings and queues data minimisation actions where applicable.
Alternative Flow 3	Patient enables Adult Mode from profile settings: system switches to simplified navigation, larger font size (minimum 18px), and reduced feature set for the session.



Business Rules

- BR-05: Consent withdrawal is recorded with timestamp and user ID for audit purposes.
- BR-06: Emergency contact information is used only during emergency support flow (UC-10) and is not shared publicly.
- BR-07: Patients can upload a profile photo; file size must not exceed 2 MB; accepted formats are JPG and PNG.

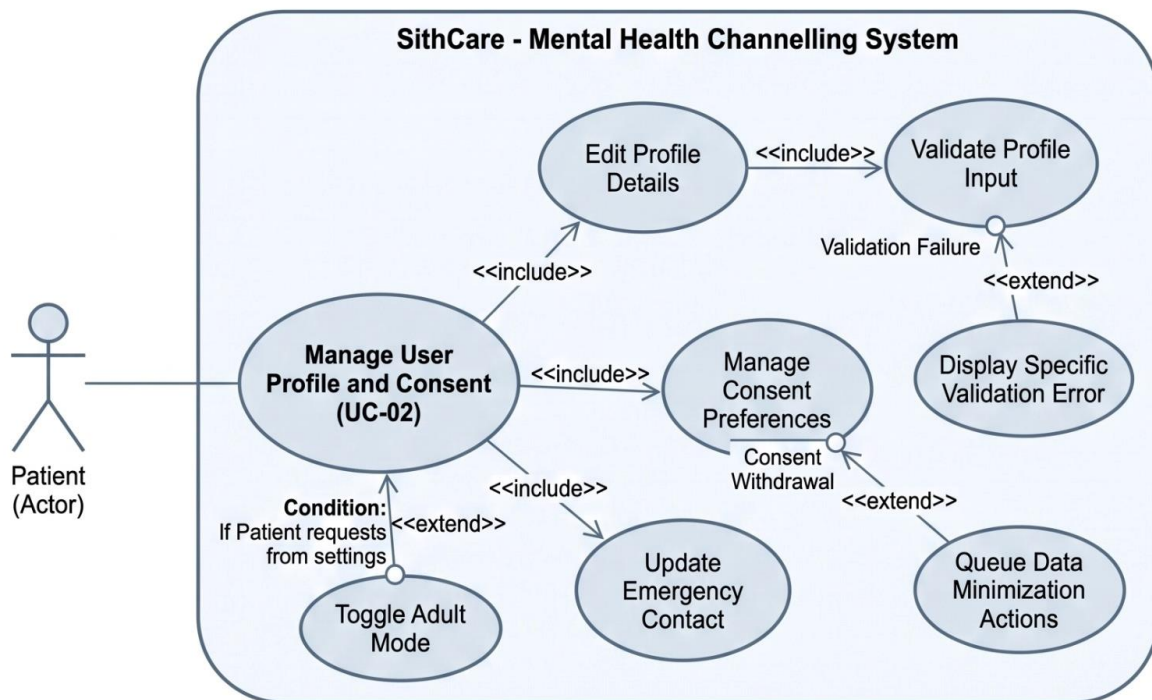


Figure 2 - UCD 02 (Manage User Profile and Content)



2.3. UC-03 – Search Mental Health Specialist

Use Case ID	UC-03
Use Case Name	Search Mental Health Specialist
Actor(s)	Patient
Preconditions	Patient is logged in. At least one verified specialist is listed on the platform.
Postconditions – Success	A filtered list of matching specialists is displayed with profile summaries and available slots.
Postconditions – Failure	No matching specialists found; system displays 'No results found' with suggestions to broaden the search criteria.
Main Flow	1. Patient navigates to 'Find a Specialist' from the dashboard or the AI chatbot recommendation. 2. Patient enters search criteria using available filters: specialist category (e.g., CBT Therapist, Psychiatrist, Counsellor), preferred language (Sinhala, Tamil, English), consultation type (online/in-person), location/district, and availability date. 3. System queries the database and returns a ranked list of matching specialists. 4. System displays each specialist with: name, photo, category, SLMC/SLPA registration number, consultation fee in LKR, languages spoken, average rating, and next available slot.
	5. Patient clicks a specialist card to view the full profile. 6. Patient reviews full profile and clicks 'Book Appointment' to proceed to UC-04.
Alternative Flow 1	At step 3: If no specialists match all filters, system relaxes lower-priority filters (e.g., consultation type) and displays results with a note: 'Showing nearest matches – some filters were broadened.'
Alternative Flow 2	At step 4: If the patient was referred via the AI chatbot (UC-14), the recommended specialist category is pre-filled in the search filter.
Alternative Flow 3	Patient selects 'View on Map' to see clinic locations pinned on an interactive map (Google Maps / OpenStreetMap integration).



Business Rules

- BR-08: Only verified specialists with status 'Active' appear in search results.
- BR09: Search results are sorted by availability (soonest slot first) by default.
- BR-10: Specialist SLMC/SLPA registration number must be verified by the Platform Admin before the profile is publicly listed (UC-15).

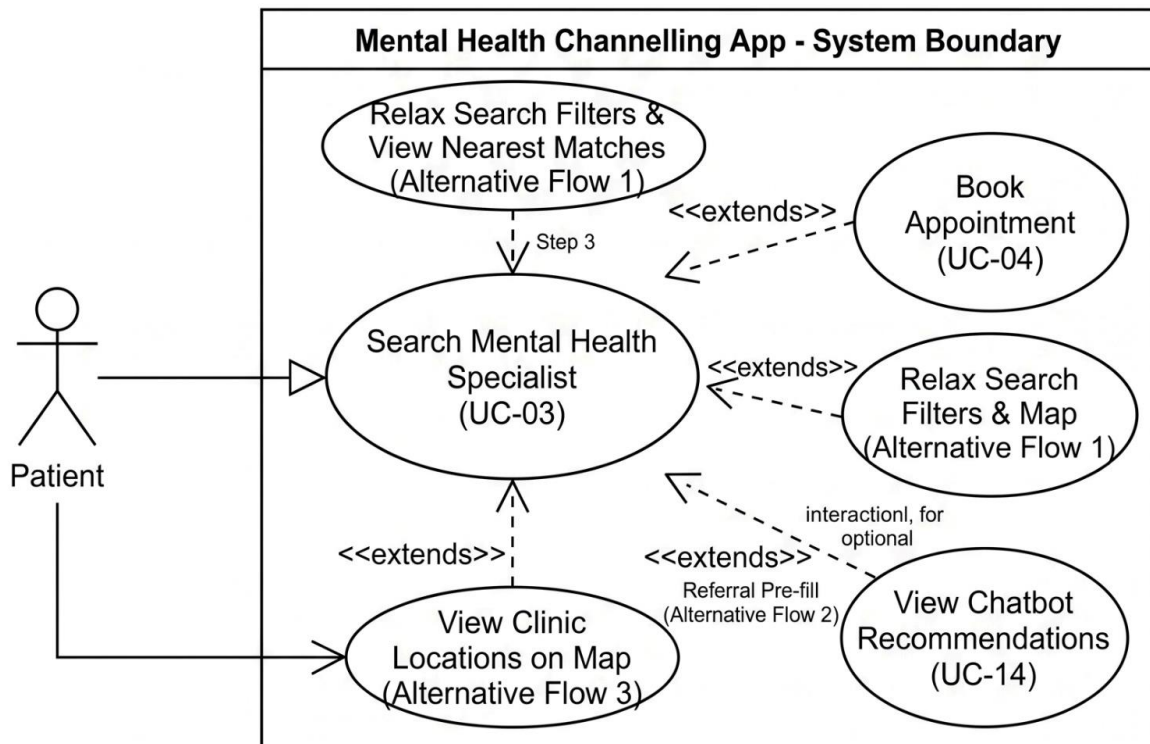


Figure 3 - UCD 03 (Search Mental Health Specialist)



2.4. UC-04 – Book Appointment

Use Case ID	UC-04
Use Case Name	Book Appointment
Actor(s)	Patient, Clinic Appointment System, Payment Gateway (PayHere), SMS/Email Service
Preconditions	Patient is registered and logged in. Patient has selected a specialist from the search results (UC-03).
Postconditions – Success	Appointment confirmed; booking reference generated; SMS and email confirmation sent; appointment appears on patient and doctor dashboards.
Postconditions – Failure	Appointment not booked; selected slot remains available; patient notified with reason.
Main Flow	1. Patient selects a specialist and clicks 'Book Appointment'. 2. System displays available time slots for the selected clinic/doctor. 3. Patient selects a date and time slot and chooses consultation type (in-person or video). 4. Patient reviews booking summary: doctor name, clinic, slot, consultation type, and fee in LKR. 5. Patient selects payment method: online (PayHere) or cash at clinic. 6. If online: patient completes PayHere checkout; system receives payment confirmation. 7. System creates appointment record with status 'Confirmed'.
	8. System generates a unique booking reference number. 9. System sends SMS and email confirmation to patient. 10. Appointment appears on both patient and doctor dashboards.
Alternative Flow 1	At step 6: If PayHere payment fails, system retains the slot for 5 minutes and prompts retry. After 2 failures, slot is released and patient is redirected to search.
Alternative Flow 2	At step 2: If no regular slots are available, system offers 'Emergency Same-Day Booking' option and displays the earliest next available slot, subject to clinic approval.
Alternative Flow 3	Patient cancels before step 7: system releases the slot and no record is created.



Business Rules

- BR-11: Appointment slots are 50 minutes by default.
- BR-12: Cancellation is free if more than 24 hours before the appointment; a cancellation fee policy applies within 24 hours.
- BR-13: Online payment receipt is generated as a PDF automatically upon payment confirmation.
- BR-14: Slot is reserved for 5 minutes during payment processing.

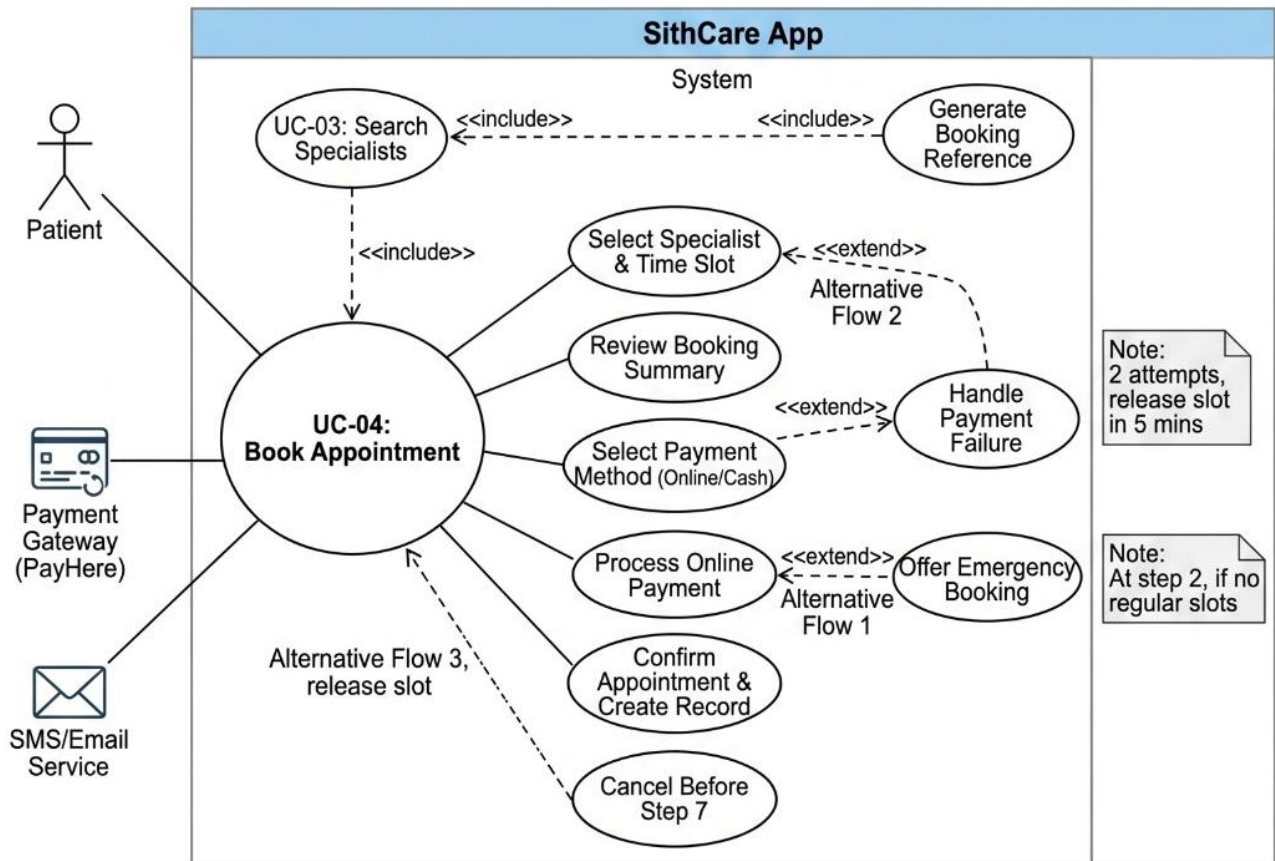
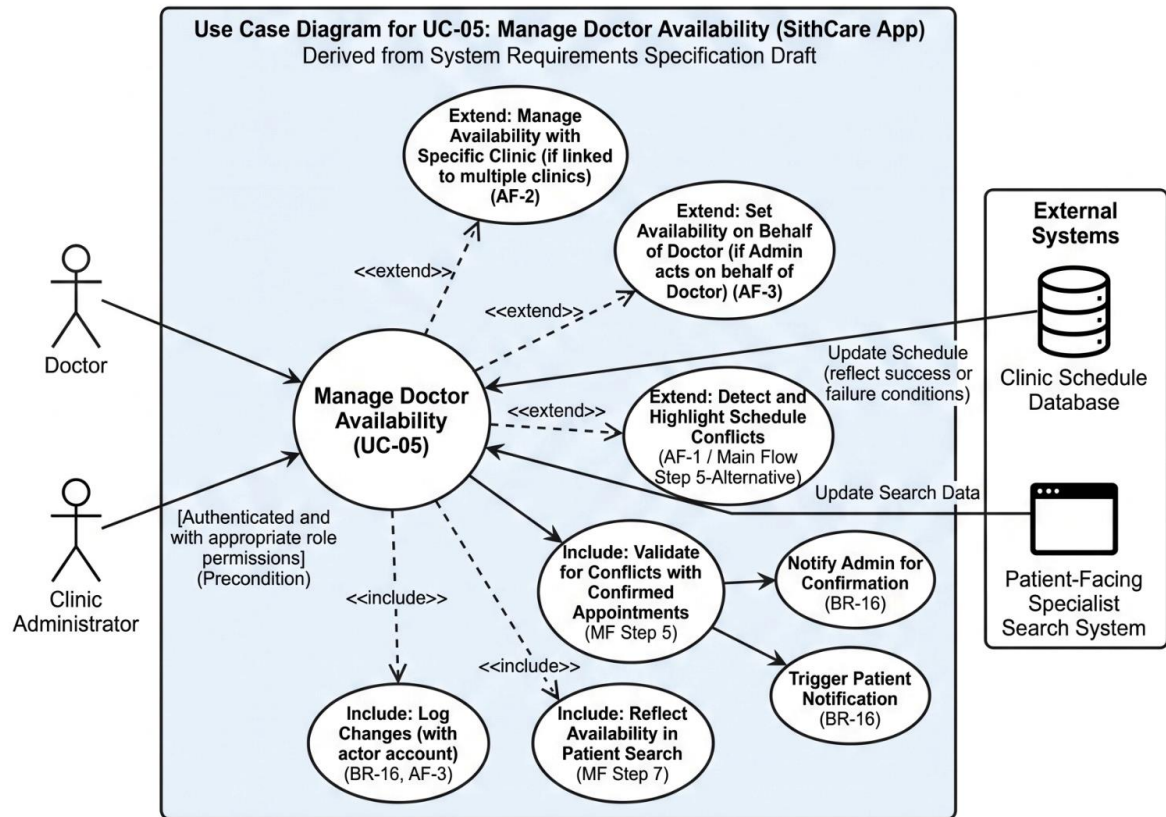


Figure 4 - UCD 04 (Book Appointment)



2.5. UC-05 – Manage Doctor Availability

Use Case ID	UC-05
Use Case Name	Manage Doctor Availability
Actor(s)	Doctor, Clinic Administrator
Preconditions	Doctor or clinic admin is authenticated and has the appropriate role permissions.
Postconditions – Success	Availability calendar is updated and reflects correctly in patient-facing specialist search.
Postconditions – Failure	Schedule conflict detected; changes not saved; user informed of the conflict.
Main Flow	1. Doctor or admin opens the availability management panel and selects a clinic/branch. 2. System displays a weekly calendar view of current availability. 3. User selects a date and defines available time slots (start time, end time, slot duration). 4. User can block specific dates/times for holidays or personal leave. 5. System validates for conflicts with existing confirmed appointments. 6. System saves the updated availability schedule. 7. Updated slots immediately become available in patient-facing search.
Alternative Flow 1	At step 5: If a conflict exists with an already-confirmed appointment, system highlights the conflict and prevents saving until resolved.
Alternative Flow 2	Doctor is linked to multiple clinics: system allows setting separate availability schedules per clinic branch.
Alternative Flow 3	Clinic admin sets availability on behalf of a doctor: system logs the change with the admin's account for audit purposes.
Business Rules	• BR-15: A doctor may be associated with up to 5 clinic branches simultaneously. • BR-16: Availability changes affecting already-confirmed appointments require explicit admin confirmation and trigger a patient notification.





2.6. UC-06 – Approve / Reschedule Appointment

Use Case ID	UC-06
Use Case Name	Approve / Reschedule Appointment
Actor(s)	Clinic Administrator, Doctor, SMS/Email Notification Service
Preconditions	One or more appointment requests with status 'Pending Approval' exist in the clinic admin dashboard.
Postconditions – Success	Appointment status updated (Confirmed/Rejected/Rescheduled); patient notified via SMS and email.
Postconditions – Failure	Action not taken; appointment remains in pending state; error displayed to admin.
Main Flow	1. Clinic admin logs in and navigates to 'Pending Appointments'. 2. System displays a list of pending appointment requests sorted by date. 3. Admin selects a request and reviews patient details, requested slot, and consultation type. 4. Admin chooses one of: Approve, Reject, or Reschedule. 5a. (Approve): System updates appointment status to Confirmed and sends confirmation to patient. 5b. (Reject): Admin enters a rejection reason. System updates status to Rejected and notifies patient with the reason and suggestion to re-book.
Alternative Flow 1	At step 5c: If patient does not confirm the rescheduled slot within 24 hours, the appointment is automatically cancelled and the patient is notified.
Alternative Flow 2	Emergency Same-Day Booking (from UC-04): admin is presented with a priority flag and must approve or reject within 30 minutes; a reminder notification is sent if not acted on within 15 minutes.
Business Rules	• BR-17: Manual approval is required only for clinics that have enabled the 'Approval Required' setting. • BR-18: Rejection reason is recorded for audit and reported to the Platform Administrator monthly.

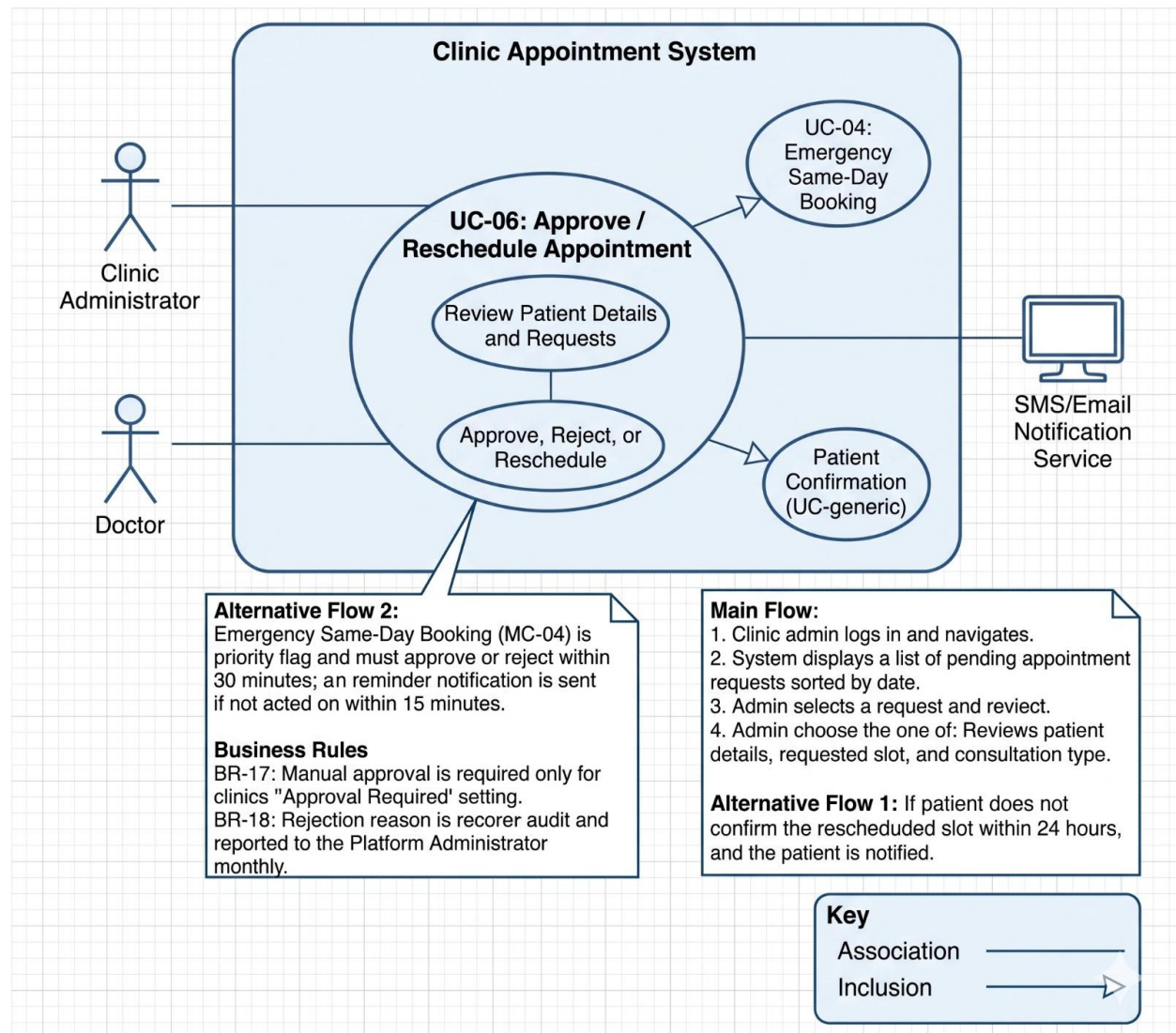


Figure 6 - UCD 06 (Approve/Reschedule Appointment)



2.7. UC-07 – Join Telemedicine Consultation

Use Case ID	UC-07
Use Case Name	Join Telemedicine Consultation
Actor(s)	Patient, Doctor, Video Consultation Service (Jitsi/Daily.co)
Preconditions	Appointment is confirmed with consultation type 'Online/Video'. Appointment time is within 15 minutes.
Postconditions – Success	Consultation session is started; both parties connected in the video room.
Postconditions – Failure	Video session fails; fallback phone call option is displayed to both parties.
Main Flow	1. System generates a unique, time-limited video room link 15 minutes before the appointment. 2. Patient opens the appointment in 'My Appointments' dashboard and clicks 'Join Session'. 3. System verifies appointment status and patient identity. 4. Patient is connected to the video room. 5. Doctor opens the appointment from the doctor dashboard and clicks 'Start Session'. 6. Doctor is connected; session status changes to 'In Progress'. 7. Upon completion, doctor clicks 'End Session'; appointment status updates to 'Completed'.
Alternative Flow 1	At step 2: Video link is expired or network failure occurs. System displays 'Unable to connect' with a one-tap phone call fallback button to call the doctor's registered phone number.
Alternative Flow 2	At step 4: Patient's device lacks camera/microphone permissions. System displays instructions for enabling permissions; audio-only fallback is offered.
Business Rules	• BR-19: Video room links expire 30 minutes after the scheduled appointment end time. • BR-20: Sessions are not recorded by the SithCare platform; recording is the doctor's responsibility if clinically required. • BR-21: One-tap phone fallback must always be available if video fails.

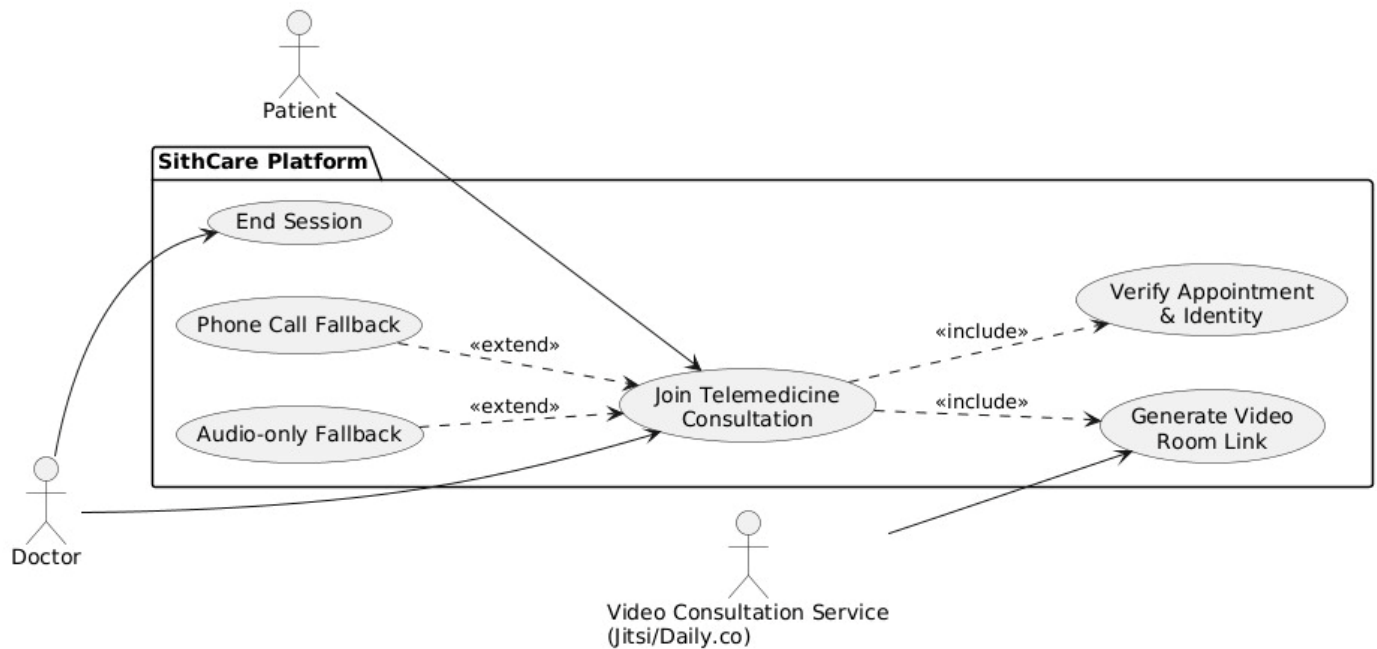


Figure 7 - UCD 07 (Join Telemedicine Consultation)

2.8. UC-08 – Process Payment and Generate Receipt

Use Case ID	UC-08
Use Case Name	Process Payment and Generate Receipt
Actor(s)	Patient, Payment Gateway (PayHere), Clinic Administrator
Preconditions	Appointment requires payment. Patient has selected a payment method (online or cash).
Postconditions – Success	Payment status updated to Paid; PDF receipt generated and available to patient and clinic admin.
Postconditions – Failure	Payment not completed; appointment status remains Pending Payment; patient notified.



Main Flow	1. Patient views payment due on the appointment summary screen. 2. Patient selects payment method: Online (PayHere) or Cash at Clinic. 3a. (Online): System redirects to PayHere checkout page with appointment details and fee in LKR. PayHere processes payment and returns status to SithCare via webhook. 3b. (Cash): Clinic admin marks the appointment as Cash Paid after receiving payment at the clinic. 4. System updates appointment/payment record to 'Paid'. 5. System generates a PDF receipt with appointment details, fee, payment method, date, and booking reference. 6. Receipt is available for download from patient dashboard and clinic admin panel.
Alternative Flow 1	At step 3a: PayHere payment fails. System retains appointment slot for 5 minutes and prompts patient to retry. After 2 failures, slot is released.
Alternative Flow 2	Refund required (appointment cancelled within refund window): System initiates refund via PayHere API; refund status tracked and patient notified of completion.
Business Rules	• BR-22: Payment card data is never stored on SithCare servers; all card handling is delegated to PayHere's PCI-DSS compliant infrastructure. • BR-23: All amounts are displayed and processed in Sri Lankan Rupees (LKR). • BR-24: PDF receipts are retained for 12 months and accessible to both patient and clinic admin.

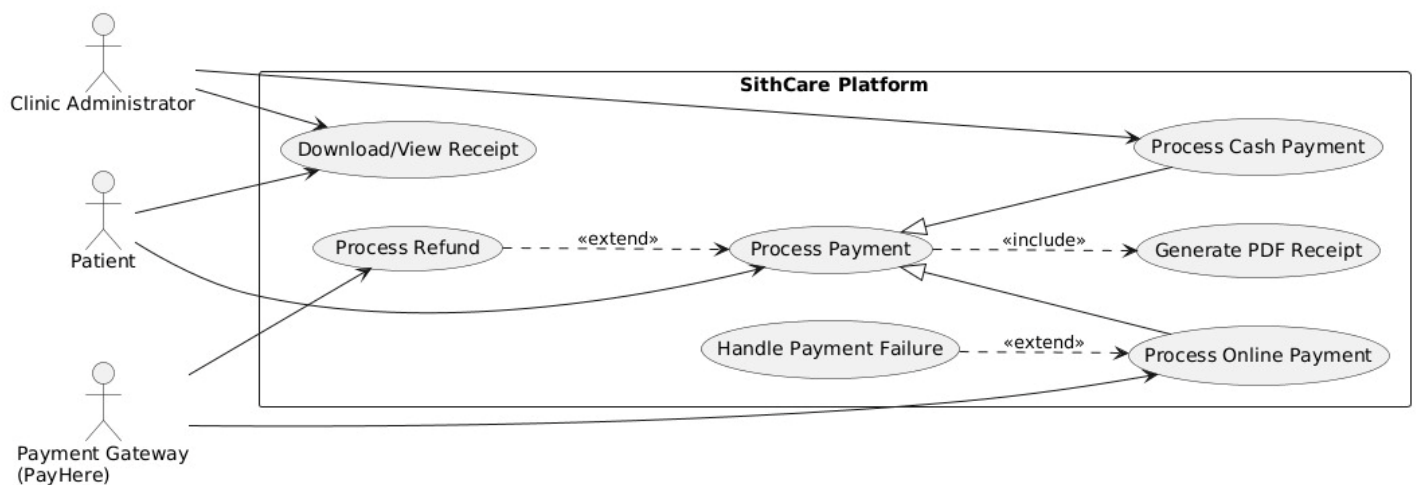


Figure 8 - UCD 08 (Process Payment & Generate Receipt)



2.9. UC-09 – Track Live Patient Queue

Use Case ID	UC-09
Use Case Name	Track Live Patient Queue
Actor(s)	Patient, Clinic Administrator
Preconditions	Patient has a confirmed in-person appointment for today with an assigned queue token. Clinic has queue tracking enabled.
Postconditions – Success	Patient can view their real-time queue position without being physically present at the clinic.
Postconditions – Failure	Queue data unavailable; static message displayed: 'Queue data temporarily unavailable. Please contact the clinic.'
Main Flow	1. Clinic admin starts the daily queue from the clinic dashboard. 2. System assigns each confirmed in-person patient a queue token number. 3. Patient logs in, navigates to 'My Appointments', and selects today's appointment. 4. System displays: current token being served, patient's token number, estimated wait time. 5. Queue tracker refreshes automatically every 60 seconds. 6. Clinic admin clicks 'Next Patient' to advance the queue. 7. System updates the display on all connected patient views in real time. 8. When patient is 3 tokens away, system sends a push/SMS notification: 'You are next soon – please head to the clinic.'
Alternative Flow 1	At step 5: If the clinic administrator pauses the queue (e.g., lunch break), a banner is shown: 'Queue is currently paused. Expected to resume at [time].'
Alternative Flow 2	Emergency slot is inserted by admin: queue is renumbered; affected patients receive an SMS notification of the delay.
Business Rules	• BR-25: Queue position is updated only by the clinic administrator. • BR-26: Patients are notified automatically when within 3 tokens of their turn. • BR-27: Queue resets automatically at the end of each clinic day.

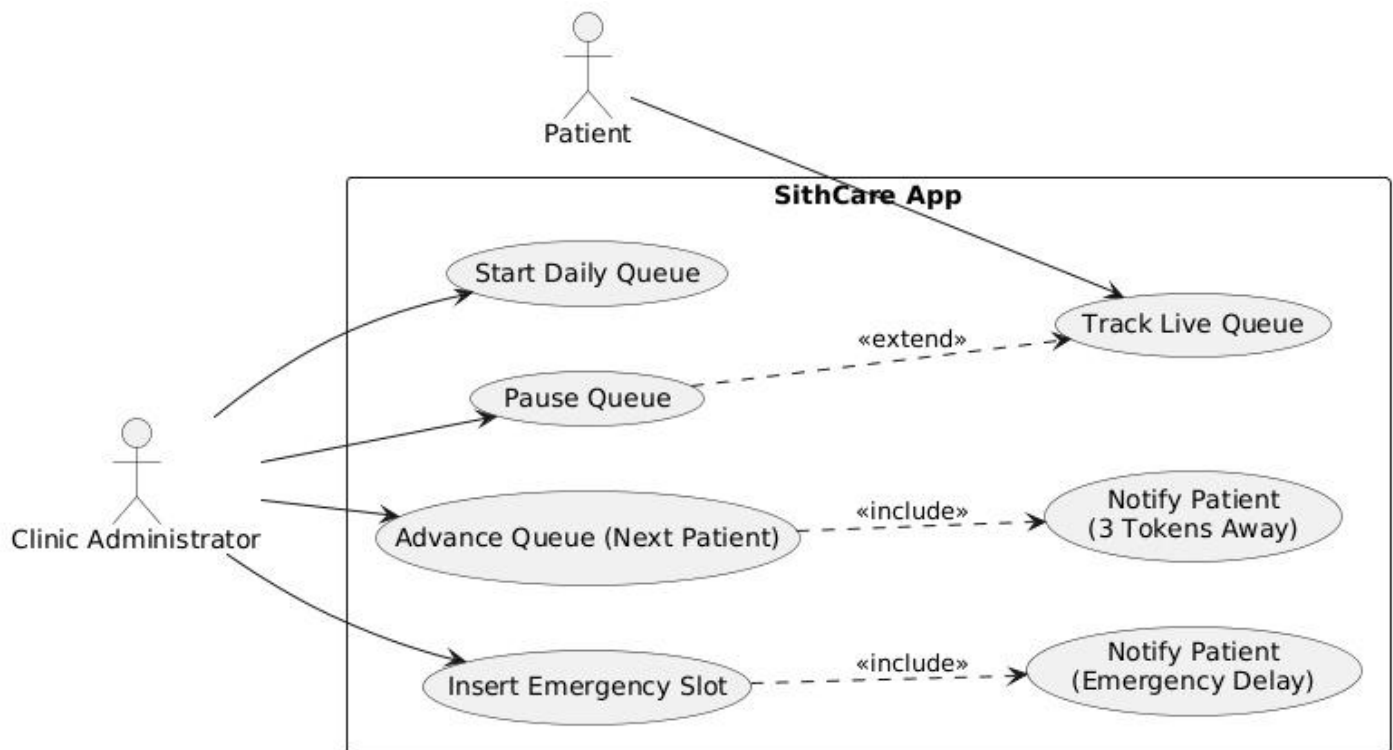


Figure 9 - UCD 09 (Track Live Patient Queue)

2.10. UC-10 – Request Emergency Support

Use Case ID	UC-10
Use Case Name	Request Emergency Support
Actor(s)	Patient, Doctor, Emergency Helpline
Preconditions	Patient is logged in and in distress. Emergency support button is visible on all screens.
Postconditions – Success	Patient is connected to assigned doctor or helpline; location shared (if consented); crisis event logged with timestamp.
Postconditions – Failure	Connection fails; patient is shown static helpline numbers with tap-to-call functionality.



Main Flow	1. Patient taps the emergency button (persistent red floating action button visible on all screens). 2. System displays: 'You are about to contact your assigned mental health professional. Your location will be shared. Confirm?' along with the disclaimer: SithCare does not replace emergency services. Dial 119 (police) or 1990 (ambulance) for immediate danger. 3. Patient taps Confirm. 4. System request's location permission from device (if not already granted). 5. System attempts to call the assigned doctor. 6. System logs the crisis event with timestamp and anonymised patient ID. 7. If doctor answers patient is connected; consented location is shared with doctor.
Alternative Flow 1	At step 3: Patient taps Cancel. System returns to previous screen. No call placed; no location shared. No crisis event logged.
Alternative Flow 2	At step 4: Location services are disabled or permission denied. System skips location sharing, proceeds with call only, and informs the doctor that location was unavailable.
Business Rules	• BR-28: Location sharing requires explicit per-session user consent. • BR-29: Crisis events are logged for safeguarding review by the Platform Administrator. • BR-30: SithCare must always display: 'SithCare is not a substitute for emergency services. Dial 119 for police or 1990 for ambulance.'

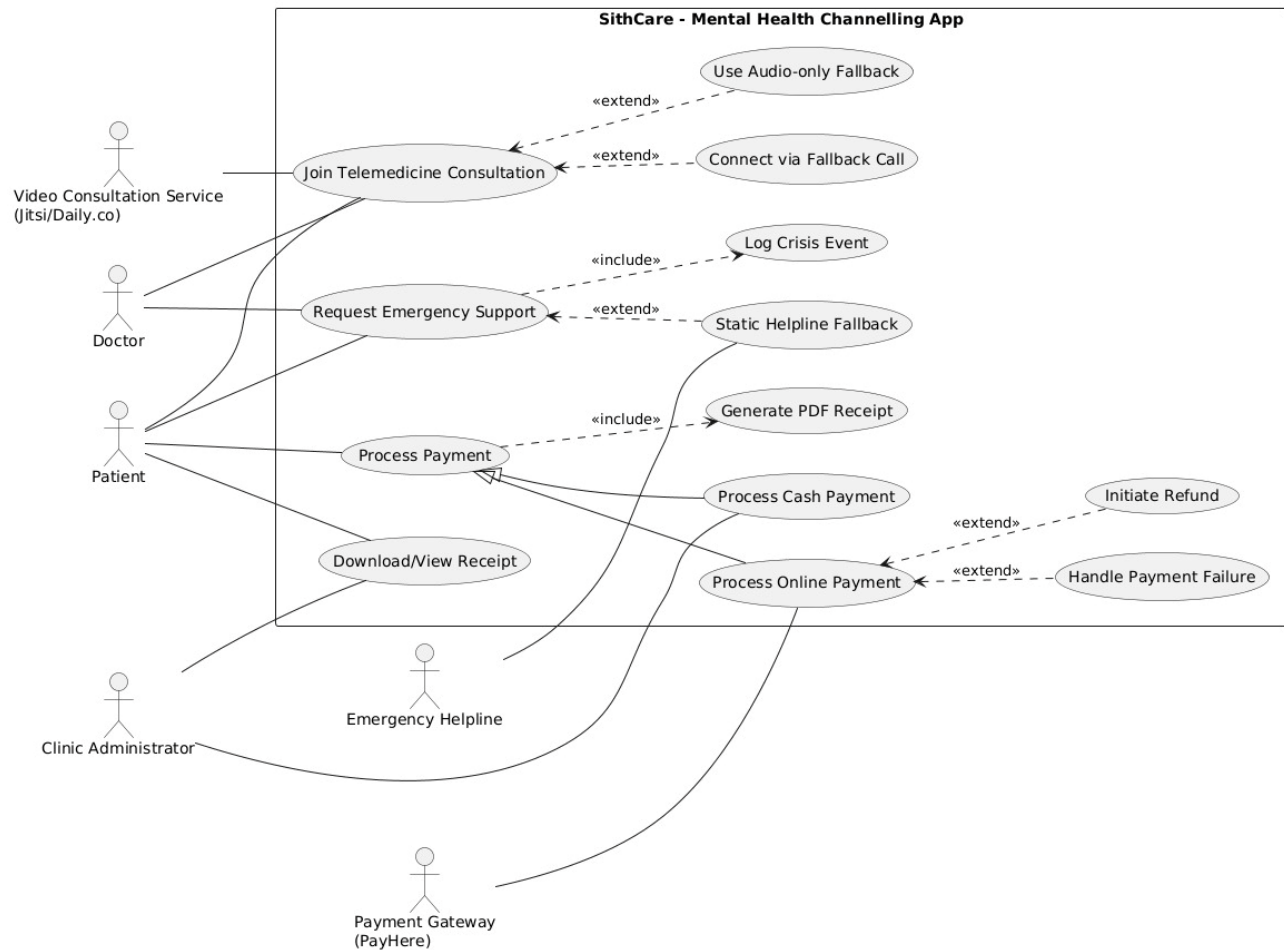


Figure 10 - UCD 10 (Request Emergency Support)

2.11. UC-11 – Use Anonymous Community Consultation

Use Case ID	UC-11
Use Case Name	Use Anonymous Community Consultation
Actor(s)	Patient, Community Moderator
Preconditions	Patient is logged in and has accepted community guidelines.
Postconditions – Success	General question is posted pseudonymously; community responses are visible; moderator can review if reported.
Postconditions – Failure	Post flagged and hidden; user informed of guideline violation.
Main Flow	1. Patient navigates to 'Community Forum' from the dashboard.



	2. System displays community safety guidelines. Patient accepts to proceed. 3. Patient submits a general mental health question using their pseudonymous platform identity. 4. System hides the real user identity & displays only the pseudonymous username. 5. Other community members (patients) can respond under their own pseudonymous identities. 6. Moderator monitors the forum and can flag, hide, or remove posts violating community guidelines.
Alternative Flow 1	Post contains private identifying information, clinical details, or emergency-level distress keywords: system warns user and recommends contacting a professional or using the emergency support feature.
Alternative Flow 2	User reports a post: moderator receives a review notification and must action within 24 hours.
Business Rules	- BR-31: Real user identity is never publicly linked to community posts. - BR-32: Community posts are not clinical advice; a disclaimer is permanently visible in the forum. - BR-33: Moderators can escalate concerning posts to Platform Admin for further review.

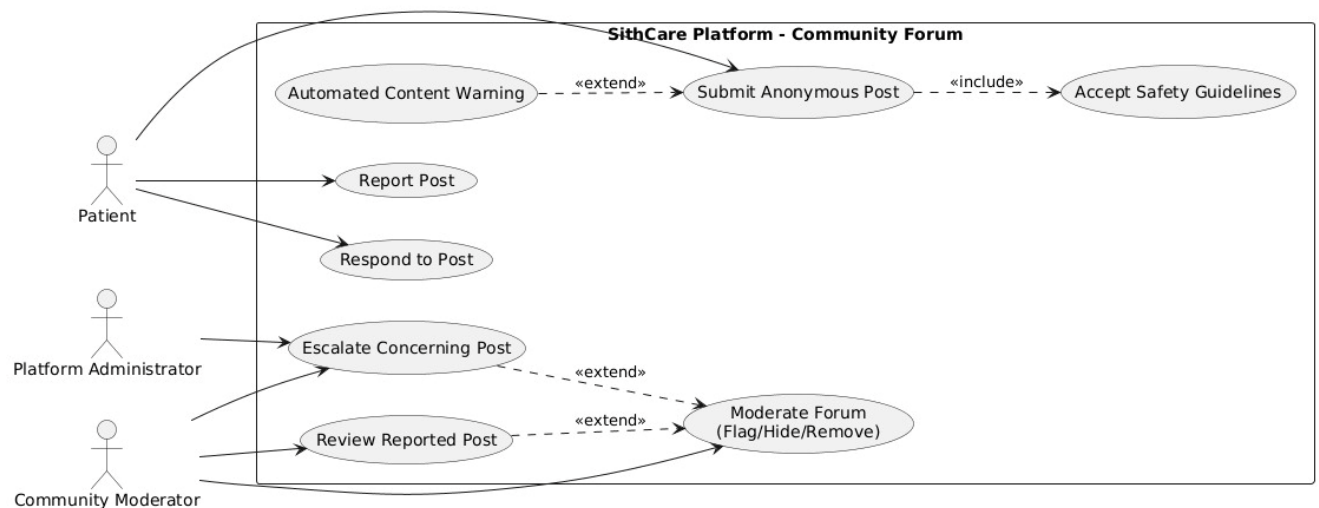


Figure 11 - UCD 11 (Use Anonymous Community Consultation)



2.12. UC-12 – Complete AI Triage Questionnaire (SithBot)

Use Case ID	UC-12
Use Case Name	Complete AI Triage Questionnaire (SithBot)
Actor(s)	Patient (or Unregistered Visitor), AI Triage Module (SithBot)
Preconditions	User has accessed the SithCare platform (login is not required for basic triage access).
Postconditions – Success	User receives a specialist category recommendation and is guided to book an appointment, access resources, or contact emergency support.
Postconditions – Failure	Chatbot cannot determine a recommendation; user is guided to contact the platform support team or browse resources manually.
Main Flow	1. User clicks 'Talk to SithBot' on the home or dashboard screen. 2. SithBot greets user and asks: 'How are you feeling today?' with quick-reply options (Anxious, Depressed, Stressed, Struggling, Not Sure). 3. User describes feelings or selects from options. 4. SithBot asks 2–4 follow-up questions based on the initial response. 5. SithBot analyses responses and suggests a specialist 6. SithBot presents options: Book Appointment Browse Resources Talk to a Volunteer Emergency Support.
	7. Medical disclaimer is displayed: 'SithBot does not provide medical diagnoses. Please consult a qualified mental health professional.'
Alternative Flow 1	At step 3: User types crisis keywords (e.g., 'suicidal', 'harm myself', 'end my life'). SithBot immediately surfaces emergency support options and helpline numbers, skipping the triage flow entirely.
Alternative Flow 2	At step 5: Chatbot confidence is below threshold. SithBot responds: 'I want to make sure you get the right support. Let me connect you with a volunteer listener or a platform support contact.'
Business Rules	• BR-34: SithBot must never use the word 'diagnose' or imply clinical assessment. • BR-35: All chatbot interactions are logged in anonymised form for safety review by Platform Admin. • BR-36: Medical disclaimer must appear on every SithBot response screen, not just the final screen.

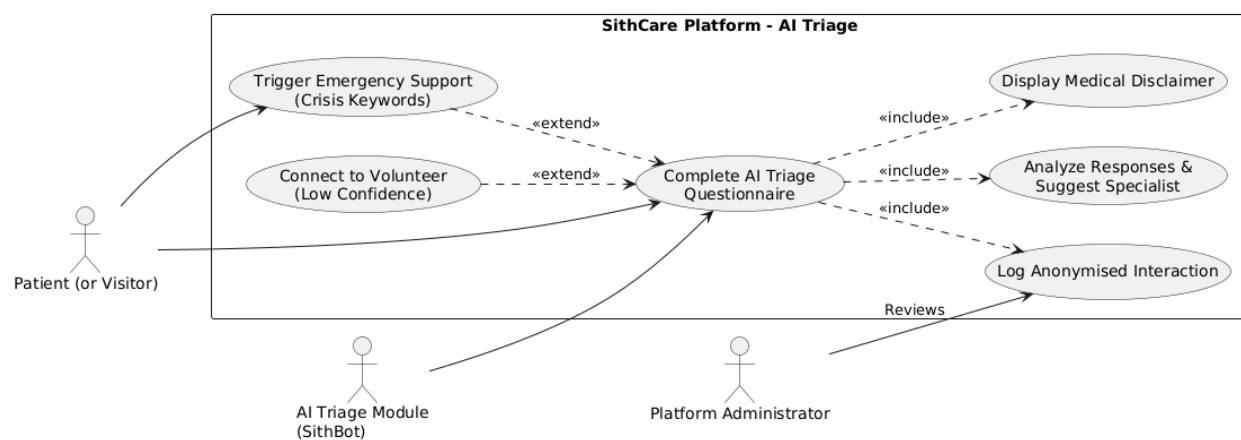


Figure 12 - UCD 12 (Complete AI Triage Questionnaire)

- End of the Document -